

AP EAMCET-2017-Engineering-26 Apr-Shift-1
Questions chemistry

121. The energy and radius of electron present in second orbit of He^+ respectively are

- (1) $-2.18 \times 10^{-18} \text{ J}$, 105.8 pm
- (2) $-4.36 \times 10^{-18} \text{ J}$, 52.9 pm
- (3) $-1.09 \times 10^{-18} \text{ J}$, 105.8 pm
- (4) $-8.72 \times 10^{-18} \text{ J}$, 211.6 pm

122. The work functions of Ag, Mg, K and Na respectively in eV are 4.3, 3.7, 2.25 and 2.30. When an electromagnetic radiation of wavelength of 300 nm is allowed to fall on these metal surfaces, the number of metals from which the electrons are ejected is
($1 \text{ eV} = 1.6022 \times 10^{-19} \text{ J}$)

- (1) 5
- (2) 4
- (3) 3
- (4) 2

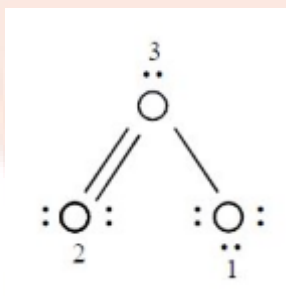
123. Assertion (A) : Na^+ and Mg^{2+} ions are isoelectronic but the ionic radius of Na^+ is greater than that of Mg^{2+}

Reason (R) : The effective nuclear charge of Na^+ ion is less than that of Mg^{2+} ion

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (3) (A) is correct but (R) is not correct.
- (4) (A) is not correct but (R) is correct.

124. The Lewis structure for O_3 molecule is given below.

The correct formal charges on oxygen atoms labelled 1, 2, 3 are respectively



- (1) $+1, 0, -1$
- (2) $-1, 0, +1$
- (3) $0, +1, -1$
- (4) $-1, -1, 0$

125. Which among the following are having, diamagnetic property?

- (a) B_2 (b) N_2 (c) O_2 (d) C_2

(1) a, b

(2) b, c

(3) a, d

(4) b, d

126. At 27°C in a 10 L flask 4.0 g of an ideal gaseous mixture containing He (molar mass 4.0 g mol^{-1}) and Ne (molar mass 20 g mol^{-1}) has a pressure of 1.23 atm. What is the mass % of neon?

($R = 0.082\text{ L atm K}^{-1}\text{ mol}^{-1}$)

(1) 84.2

(2) 25.2

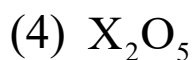
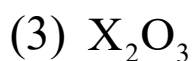
(3) 74.2

(4) 62.5

127. Two oxides of a non-metal X contains 50% and 40% of non-metal respectively. If the formula of the first oxide is XO_2 , then the formula of second oxide is

(1) XO_3

(2) X_2O



128. The enthalpy of formation ($\Delta_f H$) of methanol, formaldehyde and water $-239, -116$ and -286 kJ mol^{-1} respectively. The enthalpy change for the oxidation of methanol to formaldehyde and water in kJ is

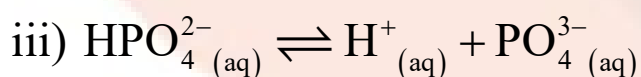
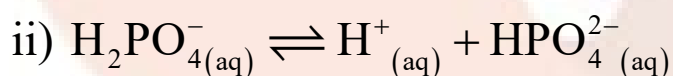
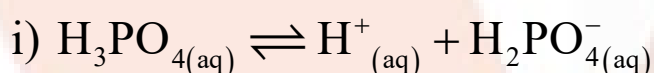
(1) -136

(2) -173

(3) -163

(4) 163

129.



The equilibrium constants for the above reactions at a certain temperature are K_1, K_2 and K_3 respectively. The equilibrium constant for the reaction $H_3PO_{4(aq)} \rightleftharpoons 3H^+_{(aq)} + PO_{4(aq)}^{3-}$ in terms of K_1, K_2 and K_3 is

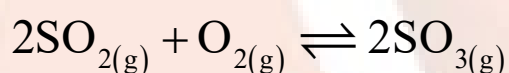
(1) $K_1 + K_2 + K_3$

(2) $\frac{K_1}{K_2 + K_3}$

(3) $\frac{K_3}{K_1 K_2}$

(4) $K_1 K_2 K_3$

130. The equilibrium constant (K_c) for the following equilibrium



at 563 K is 100. At equilibrium, the number of moles of SO_3 in the 10 litre flask is twice the number of moles of SO_2 , then the number of moles of oxygen is

(1) 0.3

(2) 0.4

(3) 0.1

(4) 0.2

131. How many milliliters of 20 volume H_2O_2 solution is needed to react completely with 500 mL of acidified 1N KMnO_4 solution?

(1) 56

(2) 140

(3) 224

(4) 280

132. Which one of the following statements is **not** correct?

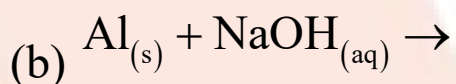
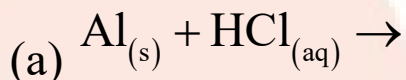
(1) Alkali metals react with water liberating oxygen gas

(2) The hydration enthalpies of alkali metals ions decrease down the group

(3) KO_2 is paramagnetic

(4) Lithium halides are somewhat Covalent in nature

133. In which of the following reactions, hydrogen is liberated?



(1) a, c

(2) a, b

(3) b, c

(4) a, b, c

134. Which one of the following statements is **not** correct?

- (1) In CO molecule, there are one 'sigma' (σ) and two "Pi" (π) bonds.
- (2) Both $[\text{SiF}_6]^{2-}$ and $[\text{SiCl}_6]^{2-}$ ions are known
- (3) In CO_2 molecule, carbon hybridisation is sp.
- (4) Fullerenes are made by heating graphite in an electric arc in the presence of argon gas.

135. Which of the following forms holes in the ozone layer?

- (1) SO_2
- (2) CO_2
- (3) CO
- (4) CF_2Cl_2

136. Identify the correct statements from the following

- a) Electromeric effect is a permanent effect
- b) Hyper conjugation is a temporary effect
- c) Fractional distillation is used to separate two liquids from a mixture if the difference in their boiling point is less
- d) Different compounds are adsorbed on an adsorbent to different extents

The correct answer is

- (1) c, d
- (2) a, b, c
- (3) b, d
- (4) b, c, d

137. The gaseous products formed at cathode (X) and anode (Y), when an aqueous solution of sodium acetate is electrolysed are

- | <u>X</u> | <u>Y</u> |
|-------------------|----------------|
| (1) C_2H_6, H_2 | CO_2 |
| <u>X</u> | <u>Y</u> |
| (2) CO_2 | C_2H_6, H_2 |
| <u>X</u> | <u>Y</u> |
| (3) H_2 | C_2H_6, CO_2 |
| <u>X</u> | <u>Y</u> |
| (4) H_2, CO_2 | C_2H_6 |

138. Identify the correct statements for a ring system to exhibit aromaticity?

- a) It must not be planar
- b) It must possess $(4n + 2)\pi$ electrons.
- c) It must be planar
- d) It must possess $4n \pi$ electrons

The correct answer is

- (1) a, b
- (2) b, c
- (3) a, d
- (4) b, d

139. An element has a body centered cubic structure with a unit cell edge length of 400 pm. Atomic mass of an element is 24 g mol^{-1} . What is the density of the element?

$$(N_A = 6 \times 10^{23} \text{ mol}^{-1})$$

- (1) 3.60 g cm^{-3}
- (2) 1.80 g cm^{-3}
- (3) 1.25 g cm^{-3}
- (4) 2.50 g cm^{-3}

140. 31 g of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) is dissolved in 600 g of water. The freezing point depression of the solution is (K_f for water is $1.86 \text{ K kg mol}^{-1}$)

- (1) 3.10 K
- (2) 4.65 K

(3) 0.77 K

(4) 1.55 K

141. Which one of the following solutions of compounds show highest osmotic pressure? (AB , AB_2 and A_2B_3 are ionic compounds)

(1) 3.0 M AB type $i = 1.6$ and temperature is 27°C

(2) 2.5 M AB_2 type $i = 2.5$ and temperature is 57°C

(3) 1.5 M A_2B_3 type $i = 4.1$ and temperature is 27°C

(4) 5.0 M urea $i = 1.0$ and temperature is 67°C

142. 20% of a first order reaction was found to be completed at 10 A.M. At 11.30 A.M. on the same day, 20% of the reaction was found to be remaining. The half life period in minutes of the reaction is

(1) 90

(2) 60

(3) 45

(4) 30

143. Same amount of electricity is passed through aqueous solutions of AgNO_3 and CuSO_4 . The number of Ag and Cu atoms deposited are x and y respectively. The correct relationship between x and y is

- (1) $x = y$
- (2) $x = 2y$
- (3) $y = 2x$
- (4) $x < y$

144. Colloidal solution of gold is in different colours like red, purple, blue and golden because of

- (1) Size difference in the particles of gold
- (2) Variable oxidation states of gold
- (3) Presence of impurities
- (4) Difference in the concentration of gold particles

145. Copper matte contains

- (1) CuO , Cu_2S
- (2) CuO , FeS
- (3) Cu_2S , FeS
- (4) Cu_2S , FeO



Here X is a gas and Y is a liquid and both are triatomic molecules. The number of electron lone pairs present on the central atoms of X and Y are respectively

- (1) 2, 1
- (2) 2, 2
- (3) 1, 2
- (4) 1, 0

147. Which of the following is used in the estimation of carbon monoxide?

- (1) Cl_2O_7
- (2) BrO_3
- (3) I_2O_4
- (4) I_2O_5

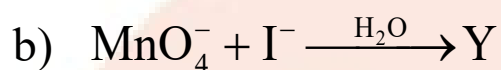
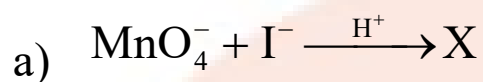
148. To prepare XeF_6 , Xe and F_2 are mixed at 573 K and 60-70 bar in the ratio of

- (1) 1:20
- (2) 20:1

(3) 1:5

(4) 5:1

149. What are X and Y in the following reactions ?



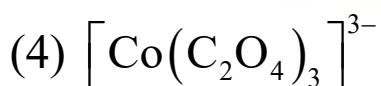
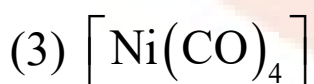
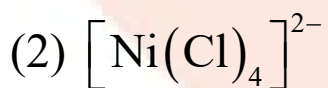
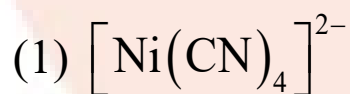
(1) IO_3^- , I_2

(2) I_2 , IO_3^-

(3) IO_3^- , IO_4^-

(4) I_2 , IO_4^-

150. Identify from the following, the diamagnetic, tetrahedral complex



151. Which one of the following is not used as an initiator in ionic polymerisation?

- (1) AlCl_3
- (2) SnCl_2
- (3) NaNH_2
- (4) $(\text{C}_6\text{H}_5\text{CO})_2\text{O}_2$

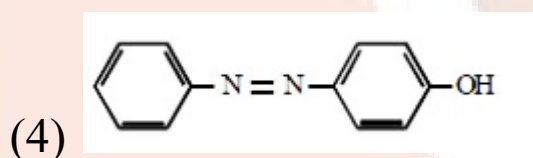
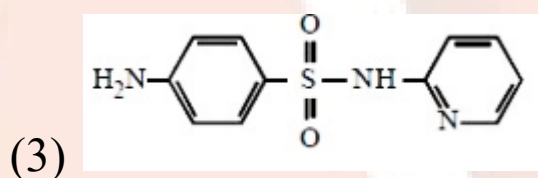
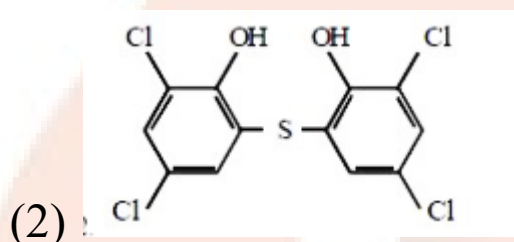
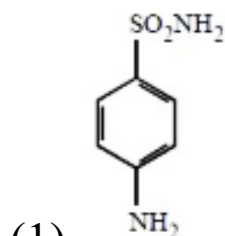
152. Identify the statements which are **not** correct from the following

- a) Carbohydrates are stored as glycogen in animals
- b) In glycylalanine, $-\overset{\text{O}}{\parallel}{\text{C}}-$ of peptide bond belongs to alanine
- c) Base – Sugar – phosphate unit is known as nucleoside
- d) Obesity is due to hypothyroidism

The correct answer is

- (1) a, c, d
- (2) a, d
- (3) b, c
- (4) b, c, d

153. Which of the following structure represents the compound, generally added to soaps to impart antiseptic properties?



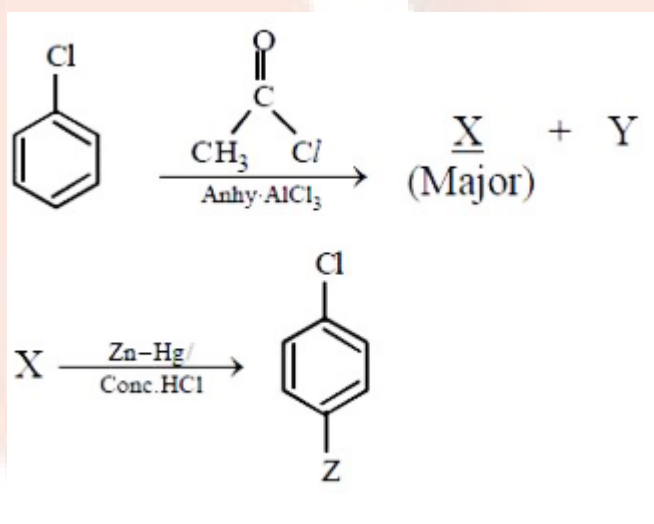
154. The drug, which was designed to prevent the interaction of histamine with the receptors present in the stomach wall is:

- (1) Equanil
- (2) Prontosil
- (3) Aspartame
- (4) Cimetidine

155. Which one of the following is more reactive towards S_N2 reaction?

- (1) $(CH_3)_3CX$
- (2) $(CH_3)_2CHX$
- (3) CH_3CH_2X
- (4) CH_3X

156. What is Z in the following sequence of reactions?

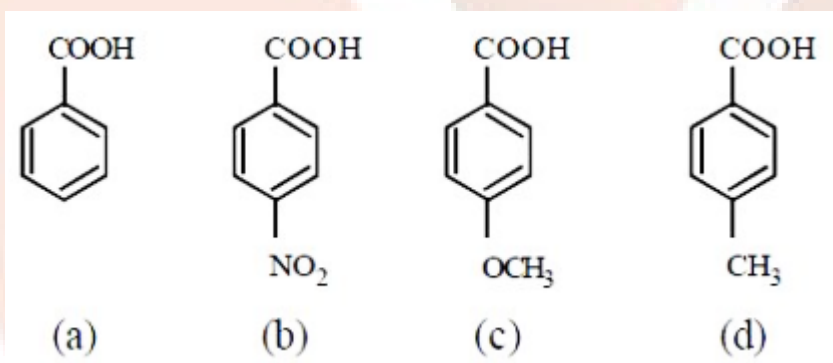


- (1) $-\text{CH}_3$
- (2) $-\text{CH}_2\text{CH}_3$
- (3) $-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{Cl}$
- (4) $-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{CH}_3$

157. Identify the statements which is **not** correct?

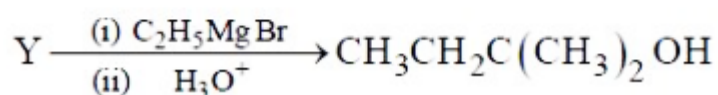
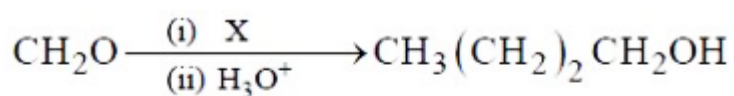
- (1) Freon 12 is manufactured by Swartz reaction
- (2) CHCl_3 is stored in closed, dark coloured bottles
- (3) Chronic exposure to CHCl_3 causes liver damage
- (4) Dehydrobromination of 2-bromopentane gives pent-1-ene as the major product

158. The increasing order of acidity of the following carboxylic acids is



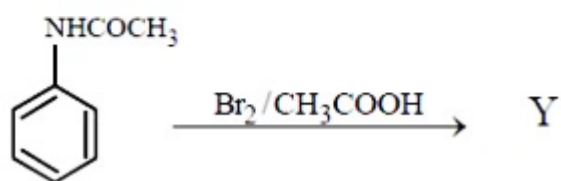
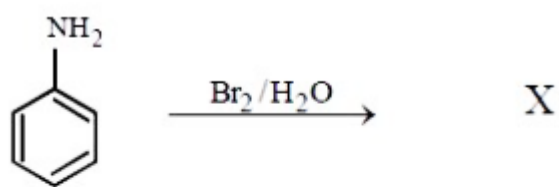
- (1) $a < b < d < c$
- (2) $c < d < a < b$
- (3) $c < d < b < a$
- (4) $b < c < d < a$

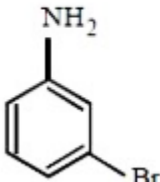
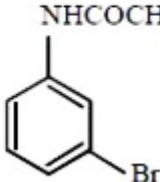
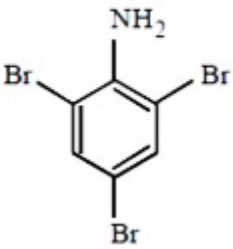
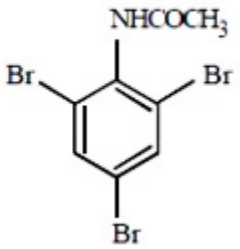
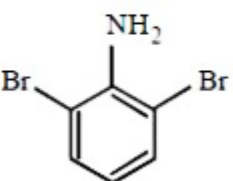
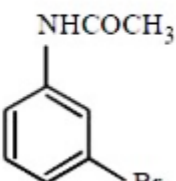
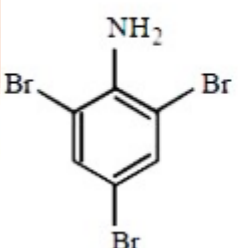
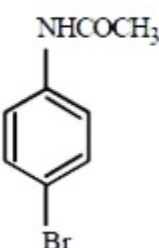
159. What are X and Y in the following reactions?



- | | <u>X</u> | <u>Y</u> |
|-----|---|---|
| (1) | $\text{CH}_3 - \underset{\text{CH}_3}{\underset{ }{\text{CH}}} - \text{MgBr}$ | $\text{C}_2\text{H}_5\text{COCH}_3$ |
| (2) | $\text{CH}_3 - \text{CH}_2 - \text{MgBr}$ | $\text{CH}_3\text{CH}_2\text{CHO}$ |
| (3) | $\text{CH}_3\text{CH}_2\text{CH}_2\text{MgBr}$ | $\text{CH}_3 - \text{CO} - \text{CH}_3$ |
| (4) | $(\text{CH}_3)_3\text{CMgBr}$ | $\text{CH}_3 - \text{CO} - \text{CH}_3$ |

160. What are X and Y in the following reactions?



- (1) X Y
-  
- (2) X Y
-  
- (3) X Y
-  
- (4) X Y
-  

AP EAMCET-2017-Engineering-26 Apr-Shift-1
Solutions – Chemistry

121. The energy is calculated as:

$$\begin{aligned}E_n &= -2.18 \times 10^{-18} \times \frac{Z^2}{n^2} \\&= -2.18 \times 10^{-18} \times \frac{4}{4} \\&= -2.18 \times 10^{-18} \text{ J}\end{aligned}$$

The radius is calculated as:

$$\begin{aligned}r_n &= 52.9 \times \frac{n^2}{Z} \\&= 52.9 \times \frac{4}{2} \\&= 105.8 \text{ pm}\end{aligned}$$

122. The energy is calculated as:

$$\begin{aligned}E &= h \frac{c}{\lambda} \\&= 6.6 \times 10^{-34} \text{ Js} \times \frac{3 \times 10^8 \text{ m/s}}{300 \times 10^{-9} \text{ m}} \\&= \frac{1.98 \times 10^{-25}}{300 \times 10^{-9} \text{ m}} \\&= 6.6 \times 10^{-19} \text{ J}\end{aligned}$$

The energy in eV is derived as:

$$1 \text{ eV} = 1.602 \times 10^{-19}$$

$$\begin{aligned}\text{Energy} &= \frac{6.6 \times 10^{-19}}{1.602 \times 10^{-19}} \\ &= 4.119 \text{ eV}\end{aligned}$$

The metals which are having less value of work function than the energy calculated, they can undergo photoelectric effect.

Therefore, metals are Mg, K and Na

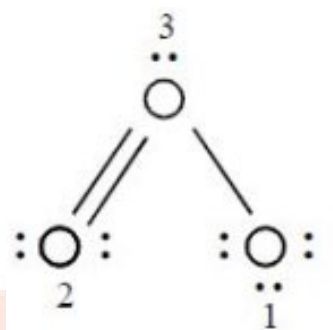
123. Since, we know that for isoelectronic species, radius is inversely proportional to effective nuclear charge.

Therefore, both Assertion (A) and Reason (R) are correct and (R) is correct explanation of (A).

124. The formal charge is calculated by using the formula shown below.

FC = Number of Valence electrons –

$$\left[\text{number of lone pair} + \frac{1}{2} \text{number of bond pair} \right]$$

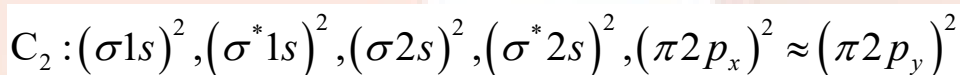
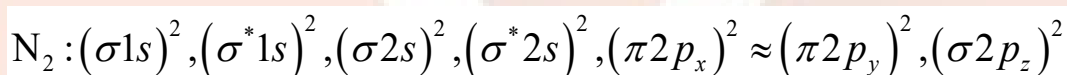


For first oxygen, formal charge is equal to $6 - [6 + 1] = -1$

For second oxygen, formal charge is equal to $6 - [4 + 2] = 0$

For third oxygen, formal charge is equal to $6 - [2 + 3] = 1$

125. The electronic configuration of the species which are having diamagnetic property (paired electrons) are shown below:



126. Total mass is $W_{\text{He}} + W_{\text{Ne}} = 4 \text{ g} \dots\dots\dots(\text{I})$

By using ideal gas equation;

$$PV = (n_{\text{He}} + n_{\text{Ne}})RT$$

$$PV = \left(\frac{W_{\text{He}}}{M_{\text{He}}} + \frac{W_{\text{Ne}}}{M_{\text{Ne}}} \right) RT$$

$$\frac{PV}{RT} = \left(\frac{W_{\text{He}}}{M_{\text{He}}} + \frac{W_{\text{Ne}}}{M_{\text{Ne}}} \right)$$

Substitute all the values.

$$\frac{1.23 \times 10}{0.082 \times 300} = \left(\frac{W_{\text{He}}}{4} + \frac{W_{\text{Ne}}}{20} \right)$$

$$50 = 25W_{\text{He}} + 5W_{\text{Ne}} \dots\dots(\text{II})$$

By subtracting equations (II) from (I), we get

$$W_{\text{He}} = 1.5$$

$$W_{\text{Ne}} = 2.5$$

Mass percentage of neon is calculated as:

$$\begin{aligned} \text{Mass \% of Ne} &= \frac{2.5 \times 100}{4} \\ &= 62.5\% \end{aligned}$$

127. The formula of first oxide is XO_2 which means one mole of X contains 2 mole of oxygen.

For second oxide, there is 40% of (X) which means there is 60% of oxygen.

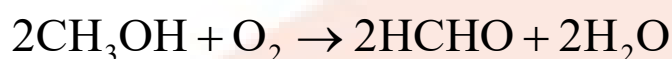
Now, for 40% of X, we have $2 \times 60\% = 80\%$ of oxygen,

The moles of oxygen for 100% of X can be calculated as:

$$\begin{aligned} \text{Moles of oxygen} &= \frac{2 \times 60}{4} \\ &= 3 \text{ moles} \end{aligned}$$

Therefore, the formula of second oxide will be XO_3 .

128. The oxidation reaction of methanol to formaldehyde and water is shown below.



The standard enthalpy change for this reaction is calculated by using the expression shown below

$$\Delta H_{\text{reaction}}^{\circ} = \sum m \cdot \Delta H_f^{\circ} (\text{products}) - \sum n \cdot \Delta H_f^{\circ} (\text{reactants})$$

$$\Delta H_{\text{reaction}}^{\circ} = \frac{1}{2} [2 \cdot \Delta H_f^{\circ} (\text{HCHO}) + 2 \cdot \Delta H_f^{\circ} (\text{H}_2\text{O})] - \frac{1}{2} [2 \cdot \Delta H_f^{\circ} (\text{CH}_3\text{OH}) + \Delta H_f^{\circ} (\text{O}_2(\text{g}))]$$

$$\begin{aligned} \Delta H_{\text{reaction}}^{\circ} &= \frac{1}{2} [2(-116) + 2(-286)] - \frac{1}{2} [2(-239) + (0)] \\ &= \frac{1}{2} [-232 - 572] - \frac{1}{2} [-478] \\ &= -163 \text{ kJ} \end{aligned}$$

129. The expression of equilibrium constant for all the given reactions are shown below.

$$K_1 = \frac{[\text{H}^+][\text{H}_2\text{PO}_4^-]}{[\text{H}_3\text{PO}_4]}$$

$$K_2 = \frac{[H^+][HPO_4^{2-}]}{[H_2PO_4^-]}$$

$$K_3 = \frac{[H^+][PO_4^{3-}]}{[HPO_4^{2-}]}$$

Now, we have to calculate value of K for the following reaction.

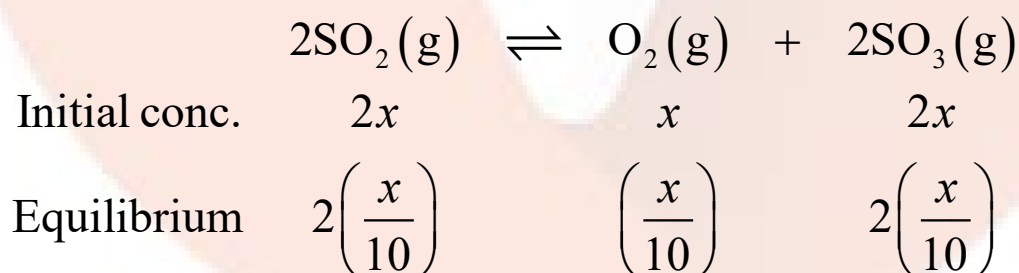


By multiplying K_1, K_2, K_3 , we get equilibrium constant of above reaction.

$$K_1 \times K_2 \times K_3 = K$$

130. Let's assume that number of moles of oxygen is 'x'

The dissociation reaction is as follows:



The equilibrium constant is calculated as:

$$K = \frac{\left(\frac{2x}{10}\right)^2}{\left(\frac{x}{10}\right)^2 \left(\frac{x}{10}\right)}$$

By simplifying the above equation, we get $x = 0.4$.

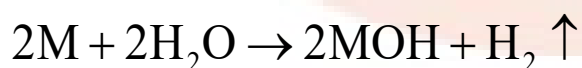
131. First calculate normality of H_2O_2 (N_2) as shown below.

$$\begin{aligned}\text{H}_2\text{O}_2 &= \frac{\text{volume of O}_2 \text{ at NTP}}{5.6} \\ &= \frac{20}{5.6} \\ &= 3.57\end{aligned}$$

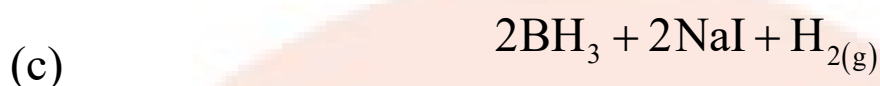
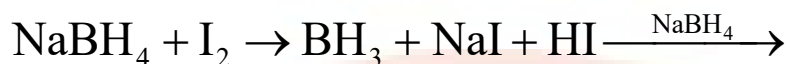
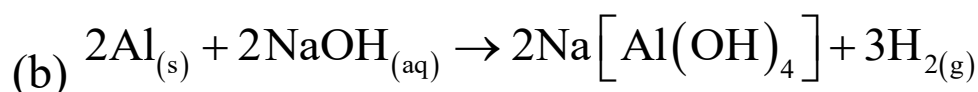
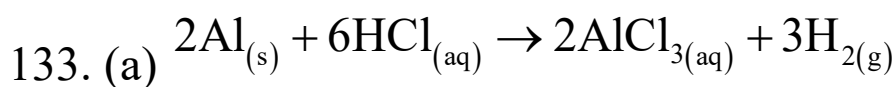
Apply the formula as follows:

$$\begin{aligned}N_1 V_1 &= N_2 V_2 \\ 1 \times 500 &= 3.57 \times V_2 \\ V_2 &= 140 \text{ mL}\end{aligned}$$

132. When alkali metals reacts with water, they form metal hydroxide and hydrogen gas is liberated as follows:



Therefore, statement (1) is not correct.



Therefore, hydrogen is liberated in all the given reactions.

134. The species $[\text{SiF}_6]^{2-}$ exists but $[\text{SiCl}_6]^{2-}$ cannot exist because the size of chlorine atom is large. Therefore, $[\text{SiCl}_6]^{2-}$ is not known.

Thus, statement (2) is not correct.

135. Chloro-fluoro carbon releases chlorine atom which reacts with ozone and produce holes in the layer of ozone.

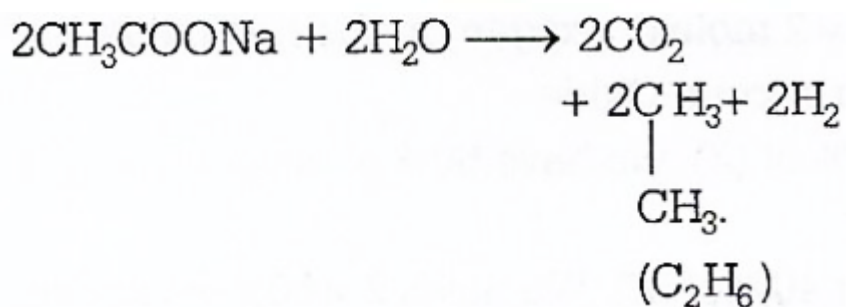
136. a) Electromeric effect is a temporary effect.

b) Hyper conjugation is a permanent effect.

c) For the separation of two liquids whose difference of boiling points is less, fractional distillation is used.

d) Different compounds are adsorbed on an adsorbent to different extents.

137. When an aqueous solution of sodium acetate is electrolysed, the reaction occurs as follows:



At anode (Y), C₂H₆ and CO₂ is formed.

At cathode (X), H₂ gas is liberated.

138. For showing the aromaticity,

The ring must possess $(4n + 2)\pi$ electrons.

The ring must be planar.

Therefore, statements (b) and (c) are correct.

139. For a body centered unit cell, the number of atoms is 2.

The density is derived by using the relation shown below.

$$d = \frac{zM}{N_A a^3}$$

$$d = \frac{2 \times 24 \text{ g/mol}}{6 \times 10^{23} \text{ mol}^{-1} \times (400 \times 10^{-10} \text{ cm})^3}$$

$$= 1.25 \text{ g/cm}^3$$

140. The change in freezing point is calculated as

$$\Delta T_f = iK_f m$$

$$= iK_f m$$

$$= \frac{i \times K_f \times W_{\text{C}_2\text{H}_6\text{O}_2} \times 1000}{M_{\text{C}_2\text{H}_6\text{O}_2} \times W_{\text{H}_2\text{O}}}$$

$$= \frac{1.86 \times 31 \times 1000}{62 \times 600}$$

$$= 1.55 \text{ K}$$

141. The osmotic pressure is calculated as:

$$\pi = iCRT$$

For 5.0 M urea,

$$\pi = 1 \times 5 \times R \times 340$$

$$= 1700R$$

For 1.5 M A_2B_3 type,

$$\pi = 4.1 \times 1.5 \times R \times 300$$

$$= 1800R$$

For 3 M AB type,

$$\begin{aligned}\pi &= 1.6 \times 3 \times R \times 300 \\ &= 1440R\end{aligned}$$

For 2.5 M AB₂ type,

$$\begin{aligned}\pi &= 2.5 \times 2.5 \times R \times 330 \\ &= 2062R\end{aligned}$$

Therefore, highest osmotic pressure is shown by the sample 2.5 M AB₂ type.

142. For a first order reaction, time can be calculated as:

$$\begin{aligned}t &= \frac{2.303}{k} \log \frac{100 - 20}{100 - 80} \\ k &= \frac{2.303}{90} \log \frac{80}{20} \\ &= 0.015 \text{ min}^{-1}\end{aligned}$$

The half-life is calculated as:

$$\begin{aligned}t_{1/2} &= \frac{0.693}{k} \\ &= \frac{0.693}{0.015} \\ &= 45 \text{ min}\end{aligned}$$

143. 1 mole of Ag⁺ is equal to $\frac{1}{2}$ mole of Cu²⁺

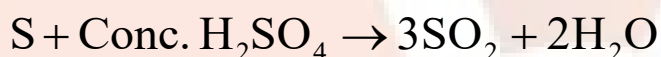
Therefore, for z moles of Ag^+ , we obtain $\frac{y}{2}$ mole of Cu^{2+}

Thus, $x = 2y$

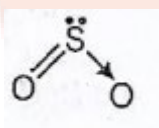
144. The difference of size in the gold particles causes the colloidal solution of gold to be in different colors like red, purple, blue and golden.

145. Copper matte is a mixture of Cu_2S and FeS

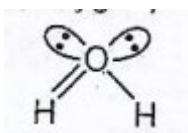
146. The reaction taking place is shown below.



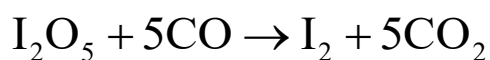
In sulphur dioxide, there is one lone pair present on central atom which is sulphur as follows:.



In water, there are two lone pairs present on central atom which is oxygen.

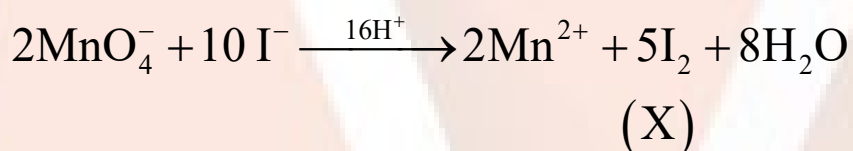


147. By using I_2O_5 , we can estimate carbon monoxide as follows:

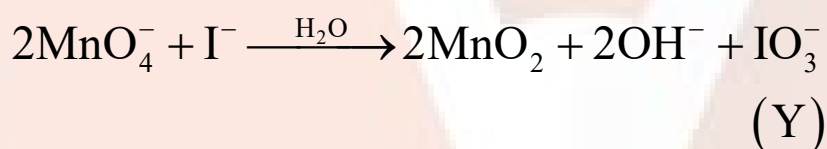


148. By taking xenon and fluorine in the ratio of 1:20 and by applying pressure of 70 bar at 573 K temperature, xenon hexafluoride is formed.

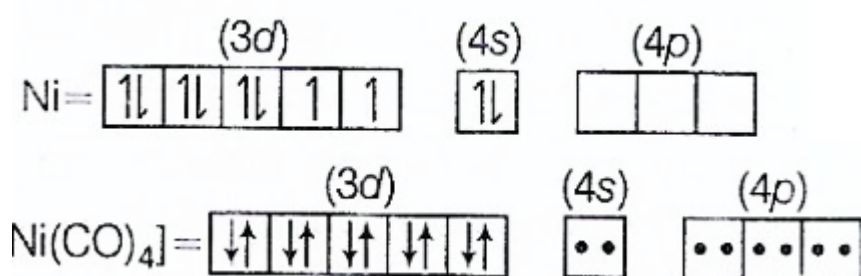
149. a) The product (X) formed is as follows:



b) The product (Y) formed is as follows:



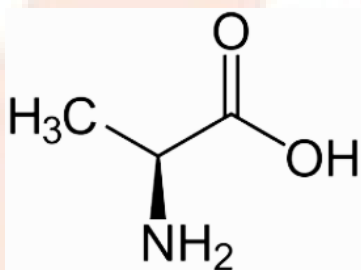
150. The tetrahedral complex is formed by $[\text{Ni}(\text{CO})_4]$ as CO is a strong ligand which causes pairing of unpaired electrons.



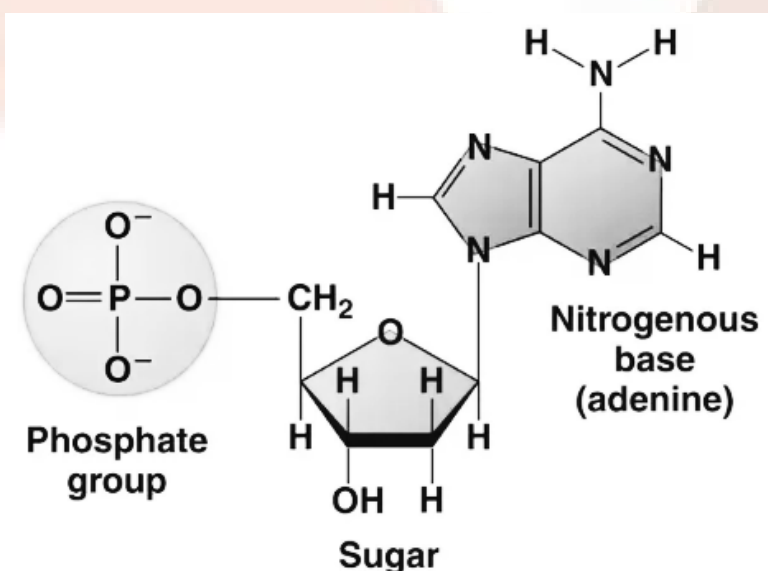
Hence, the complex is diamagnetic in nature.

151. $(\text{C}_6\text{H}_5\text{CO})_2\text{O}_2$ cannot be used as an initiator in ionic polymerization as it is a covalent compound and will not produce any free radical for chain initiation process.

152. b) The basic structure of alanine is

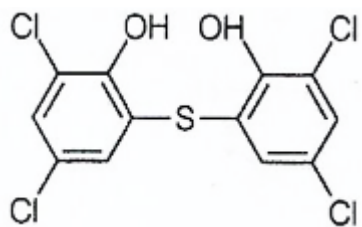


c) Nucleotide contains Base-Sugar and phosphate unit as shown below.

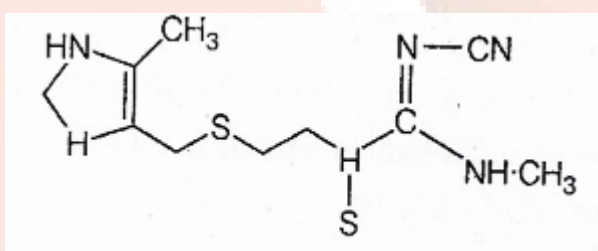


Therefore, statements (b) and (c) are incorrect.

153. The following structure is generally used to impart antiseptic properties in soap.

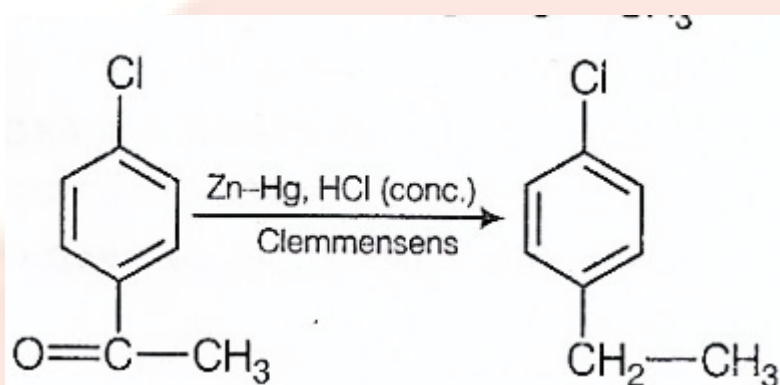
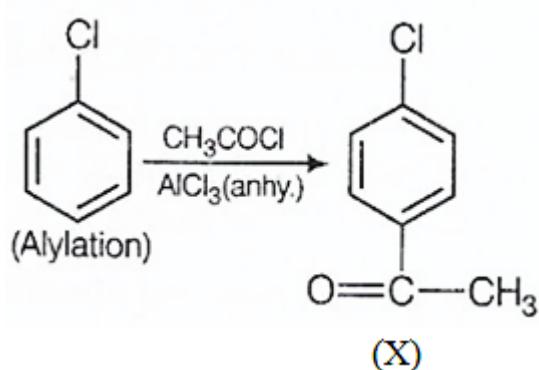


154. The drug which was discovered to avoid the interaction of histamine with the receptors present in the stomach wall is cimetidine. Its structure is shown below.



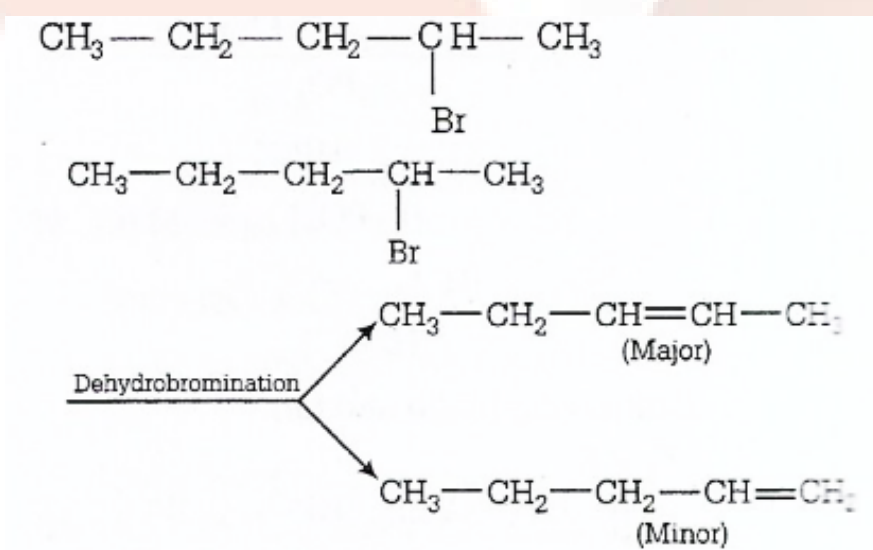
155. The species which is less bulkier is more reactive towards S_N2 reaction. Among the given species, CH_3X is least bulkier alkyl halide, therefore, its rate of reaction will be faster towards S_N2 reaction.

156. The product (Z) formed is shown below.



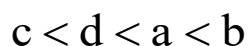
Therefore, Z is CH_2CH_3

157. Dehydrobromination of 2-bromopentane produce pent-2-ene as the major product.

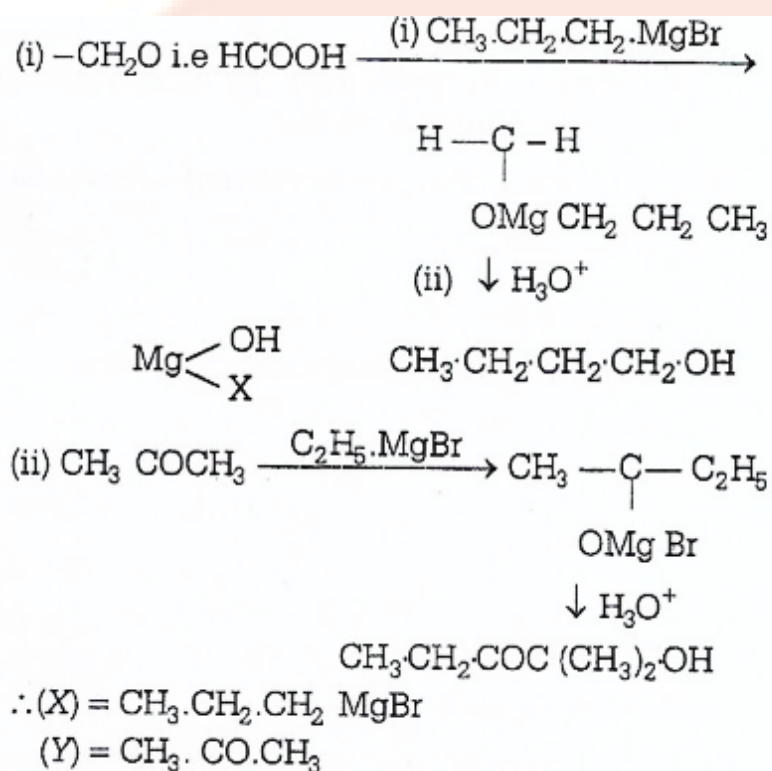


Therefore, statement (4) is not correct.

158. The increasing order of acidity is as follows:



159. The reaction is shown below.



160. The products (X) and (Y) formed are shown below.

